

## Unit 9 Progress Check: FRQ

Name \_\_\_\_\_

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to  $\text{H}_2\text{O}$ .

| Substance               | $\Delta G^\circ_f$ at 298 K (kJ/mol) |
|-------------------------|--------------------------------------|
| $\text{H}_2\text{O}(l)$ | -237.2                               |
| $\text{H}_2\text{O}(g)$ | -228.4                               |

- (a) Using the information in the table above, determine the value of  $\Delta G^\circ$  at 298 K for the process represented by the equation  $\text{H}_2\text{O}(l) \rightleftharpoons \text{H}_2\text{O}(g)$ .



Please respond on separate paper, following directions from your teacher.

- (b) Considering your answer to part (a), indicate whether the process is thermodynamically favorable at 298 K. Justify your answer.



Please respond on separate paper, following directions from your teacher.

- (c) Considering your answer to part (b), explain why  $\text{H}_2\text{O}(l)$  has a measurable equilibrium vapor pressure at 298 K.



Please respond on separate paper, following directions from your teacher.

Water vapor can be produced in two different processes, as represented below.

| Process 1                                                        | Process 2                                                        |
|------------------------------------------------------------------|------------------------------------------------------------------|
| $\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(g)$ | $\text{H}_2\text{O}(l) \rightleftharpoons \text{H}_2\text{O}(g)$ |

- (d) In terms of concepts of entropy and the particle-level structure of the different phases of water, explain why the change in entropy,  $\Delta S$ , is greater for process 1 than for process 2.

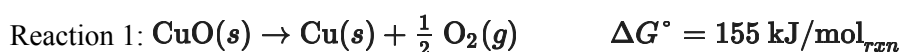


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Please respond on separate paper, following directions from your teacher.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A chemist wants to produce  $\text{Cu}(s)$  from a sample of pure  $\text{CuO}(s)$  according to reaction 1, represented by the equation above.

- (a) Using the data in the following table, calculate the value of the standard entropy change,  $\Delta S^\circ$ , for the reaction.

| Substance       | Absolute Entropy at 298 K ( $\text{J K}^{-1} \text{mol}^{-1}$ ) |
|-----------------|-----------------------------------------------------------------|
| $\text{Cu}(s)$  | 33                                                              |
| $\text{O}_2(g)$ | 205                                                             |
| $\text{CuO}(s)$ | 43                                                              |



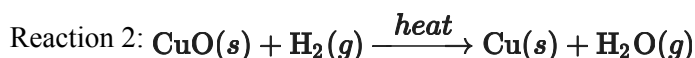
Please respond on separate paper, following directions from your teacher.

- (b) Given that  $\Delta G^\circ$  for reaction 1 is positive ( $155 \text{ kJ/mol}_{\text{rxn}}$ ), what must be true about the sign of  $\Delta H^\circ$  for the reaction? Justify your answer.



Please respond on separate paper, following directions from your teacher.

The decomposition of  $\text{CuO}(s)$  into  $\text{Cu}(s)$  and  $\text{O}_2(g)$  is not a thermodynamically favored reaction. However, to produce  $\text{Cu}(s)$  from  $\text{CuO}(s)$ , the chemist decides to pass  $\text{H}_2(g)$  gas over the  $\text{CuO}(s)$  as it is heated strongly, as represented below.



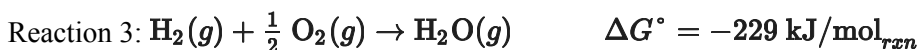
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(c) Reaction 2 is what type of chemical reaction? Justify your answer.



Please respond on separate paper, following directions from your teacher.

A third chemical reaction is represented below.



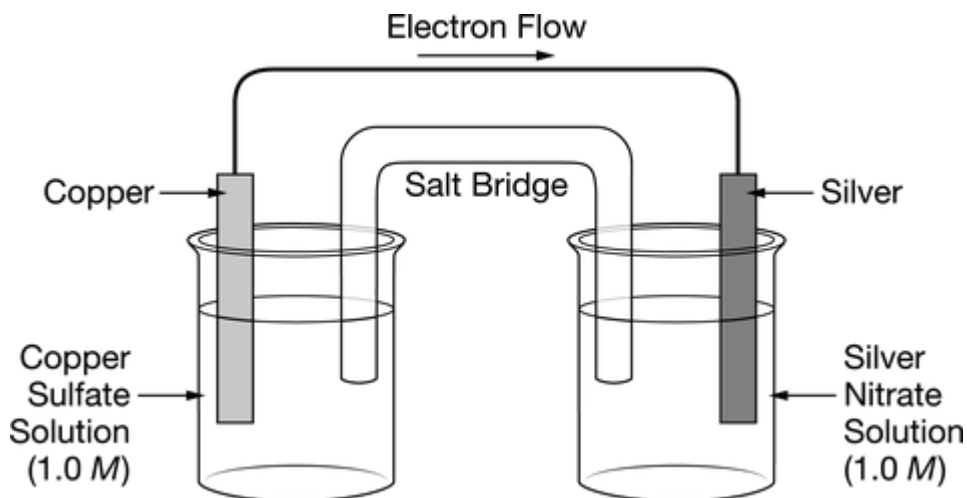
(d) Show how a combination of reaction 1 and reaction 3 can be used to produce reaction 2.



Please respond on separate paper, following directions from your teacher.

(e) Determine the value of  $\Delta G^\circ$  for reaction 2.

The chemist takes the  $\text{Cu}(s)$  produced from reaction 2 and uses it to make an electrode in a galvanic cell, as shown in the following diagram.



Please respond on separate paper, following directions from your teacher.

(f) Using the reduction half-reactions in the table below, write a balanced net-ionic equation representing the overall reaction that takes place as the cell operates.

| Half-Reaction                                         | $E^\circ$ (volts) |
|-------------------------------------------------------|-------------------|
| $\text{Cu}^{2+}(aq) + 2 e^- \rightarrow \text{Cu}(s)$ | 0.337             |



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| Half-Reaction                                    | $E^\circ$ (volts) |
|--------------------------------------------------|-------------------|
| $\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$ | 0.800             |



Please respond on separate paper, following directions from your teacher.

(g) Determine the value of  $E^\circ$  for the cell.



Please respond on separate paper, following directions from your teacher.

(h) Determine the value of  $\Delta G^\circ$  for the cell reaction.



Please respond on separate paper, following directions from your teacher.